

Smart Metro Ticket Booking System

Project Design Phase

Date	
Team ID	
Project Name	Smart Metro Ticket Booking System
Maximum Marks	4 Marks

3. Problem-Solution Fit

This document establishes the fit between the pain points experienced by each persona under the current manual, counter-based metro ticketing process and the capabilities the proposed ServiceNow solution delivers to address them.

3.1 Problem Statement

Today, metro ticket purchase relies heavily on physical counters and cash handling. Passengers face long queues during peak commute hours, limited digital payment acceptance, and no way to review past bookings. Station Managers update fare and station information manually and validate tickets without a real-time, auditable check. Metro Operations lacks consolidated visibility into ticket volume, revenue split, and peak-time demand, making schedule and resource planning reactive rather than data-driven. IT Admins cannot change fare rules or catalog configuration without raising a development request, slowing down routine business changes.

3.2 Problem — Solution Fit Mapping

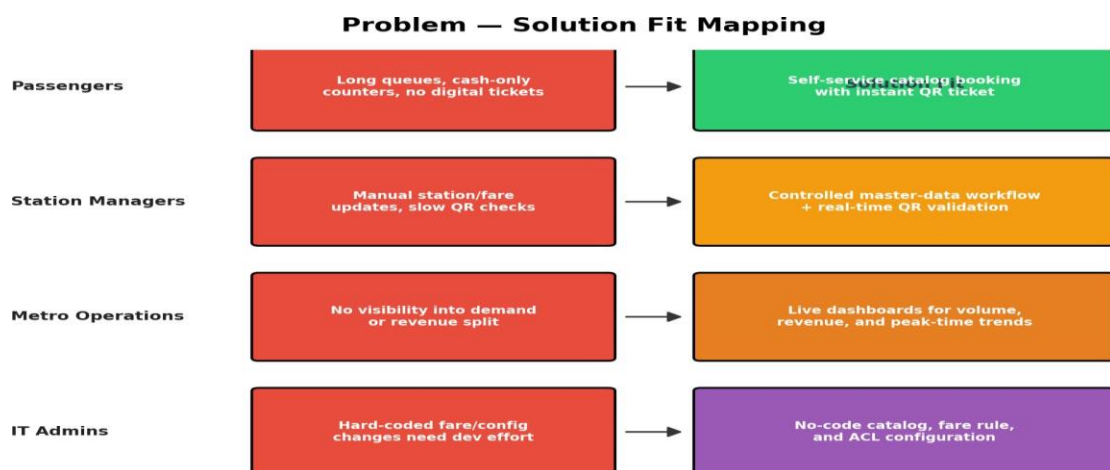


Figure 5.1 — Mapping of persona-level pain points to the proposed solution fit

Persona	Problem / Pain Point	Solution Fit
Passengers	Long queues, cash-only counters, no digital ticket or booking history.	Self-service Service Catalog booking with instant QR ticket delivery and a viewable booking history.
Station Managers	Manual station/fare updates and slow, unverifiable ticket checks at the gate.	Controlled, approval-based master-data workflow plus real-time QR validation within 2 seconds.
Metro Operations Team	No consolidated view of ticket volume, revenue, or peak travel patterns.	Performance Analytics dashboards surfacing usage, digital-vs-cash revenue split, and peak-time trends.
IT Admins	Fare and catalog changes require code deployment, slowing routine updates.	No-code configuration of catalog variables, fare rules, and ACLs directly in the ServiceNow UI.

3.3 Why Now (Market & Timing Fit)

- Rising smartphone and UPI/digital-wallet penetration means most passengers already carry the means to pay and receive tickets digitally.
- Post-pandemic commuter expectations favor contactless, counter-free ticketing over cash handling.
- The metro network already plans to deploy QR-based Automated Fare Collection (AFC) gate hardware, making this the natural moment to digitize the booking front end.